NTPC BUSINESS RESPONSIBILITY REPORT (2021-22)

Section A : General Information About Your Company

1. **CIN (Corporate identity number)** L40101DL1975GOI007966
2. **Name of the company** NTPC LIMITED.
3. **Registered address** NTPC Bhawan, SCOPE Complex, 7, Institutional Area, Lodhi Road, New Delhi-110003
4. **Website** www.ntpc.co.in
5. **Email id** csntpc@ntpc.co.in
6. **FY reported** 2021-22
7. **Sector that company is engaged in:** Power
8. **Product/services that the company manufacturers /provides (as in balance sheet):**
   i. Generation of Electricity
   ii. Consultancy
   iii. Coal Mining
   iv. Energy Trading
9. **Total number of location where business activity is undertaken by the company:**

Sl. No.	Location	Nos. of Project	Remarks
1	National	69	26 Thermal, 7 combined cycle gas/liquid fuel, 5 Hydro, 8 Coal site, 1 Wind, and 13 solar Power Projects. Further, it has 11 Thermal, 1 Coal site and 1 gas station, owned by joint ventures or subsidiaries of your Company.
2	International	2	i. Trincomalee Power Project, Srilanka ii. Bangladesh- India Friendship Power Company (Pvt) Ltd. : Maitree Super Thermal Power Plant, Rampal, Bangladesh

10. **Markets served by the company:** National & International

Section B : Financial details of the company

1	Paid up capital (₹ crore)	9,696.67
2	Total Turnover(Gross) (₹ crore)	1,15,021.82
3	Total profit after taxes (₹ crore)	16,111.42
4	Total spending on CSR and Sustainable Development (SD) as % of PAT	Your Company has been consistently spending more than the mandated CSR spend of 2%. During the FY 2021-22 your Company has spent an amount of ₹ 356.72 (2.21%) crore against the mandated CSR expenditure of ₹ 281.80 crore. Out of the said CSR Expenditure, your Company, in alignment with the guidelines issued by DPE on common Annual theme for CPSEs of "Health & Nutrition" for FY 2021-22, has spent an amount of ₹ 198.95 crore i.e. 70.60% which is more than the prescribed 60% on Annual Theme.
5	List of activities in which expenditure in 4 above has been incurred	Broad areas of the activities: - Eradicating Hunger and Poverty - Education & skill development - Health care & Sanitation - Rural Development - Women Empowerment and other Economically backward sections - Art, Culture and Sports - Environmental Sustainability - PMCARES Fund - Disaster management including relief rehabilitation and reconstruction activities

Section C: Other Details

Subsidiaries: The Company has the following 12 Subsidiary Companies as on 31.03.2022:

i. NTPC Electric Supply Company Limited (NESCL)

ii. NTPC Vidyut Vyapar Nigam Limited (NVVN)

iii. Kanti Bijlee Utpadan Nigam Limited (KANTI)

iv. Bhartiya Rail Bijlee Company Limited (BRBCL)

v. Patratu Vidyut Utpadan Nigam Limited (PVUNL)

vi. Nabinagar Power Generating Co. Ltd. (NPGCL)

vii. THDC India Limited (THDCL)

viii. North Eastern Electric Power Corporation Limited (NEEPCO)

ix. NTPC Mining Limited (NML)

x. NTPC EDMC Waste Solutions Private Limited (NEWS)

xi. NTPC Renewable Energy Limited (NREL)

xii. Ratnagiri Gas and Power Private Limited (RGPPL)

The Business Responsibility Initiatives of the parent company are applicable to the subsidiary companies also. However, none of the entities that the Company does business with, participate in the BR initiatives of the Company.

Section D: BR information

1. Individual Directors responsible for implementation of the BR policy / policies

Principle No.	Description	Policy / Policies	Director(s) Responsible
Principle 1 (P1)	Businesses should conduct and govern themselves with Ethics, Transparency and Accountability.	1. Code of Conduct* 2. Core Values 3. Fraud Prevention Policy 4. CDA Rules 5. Whistle Blower Policy 6. Internal code of conduct for prevention of insider trading 7. Code of Corporate Fair Disclosure Practices for prevention of insider trading 8. Related Party Transaction Policy 9. Policy for determination of materiality of events or information for disclosure 10. Policy on maintenance & preservation of documents 11. Policy for Determining Material Subsidiaries 12. Training Policy for Directors of NTPC	All Directors & Chief Vigilance Officer
Principle 2 (P2)	Businesses should provide goods and services that are safe and contribute to sustainability throughout their life cycle.	1. Safety Policy 2. NTPC Policy for CSR and Sustainability 3. Ash Policy 4. Sustainable Supply Chain Policy	Shri Ramesh Babu V. DIN: 08736805 Designation: Director (Operations) Shri D K Patel DIN: 08695490 Designation: Director (HR)
Principle 3 (P3)	Businesses should promote the well-being of all employees.	1. Human Resource (HR) Policies 2. Placement and Transfer Policy 3. Human Rights Policy	Shri D K Patel DIN: 08695490 Designation: Director (HR)

Principle No.	Description	Policy / Policies	Director(s) Responsible
Principle 4 (P4)	Businesses should respect the interests of, and be responsive towards all stakeholders, especially those who are disadvantaged, vulnerable and marginalized.	1. R&R Policy 2. NTPC Policy for CSR and Sustainability 3. Initial Community Development (ICD) Policy 4. Human Rights Policy	Shri D K Patel DIN: 08695490 Designation: Director (HR)
Principle 5 (P5)	Businesses should respect and promote human rights.	1. Human Rights Policy	Shri D K Patel DIN: 08695490 Designation: Director (HR)
Principle 6 (P6)	Businesses should respect, protect and make efforts to restore the environment.	1. Environment Policy 2. NTPC Policy for CSR and Sustainability 3. Biodiversity Policy 4. E-waste Policy 5. Water Policy 6. Rainwater Harvesting Policy 7. Integrated Plastic Management Policy	Shri Ramesh Babu V. DIN: 08736805 Designation: Director (Operations) Shri D K Patel DIN: 08695490 Designation: Director (HR)
Principle 7 (P7)	Businesses, when engaged in influencing public and regulatory policy, should do so in a responsible manner.	1. Code of Conduct* 2. Core Values 3. Commercial – Systems & Procedures	All Directors
Principle 8 (P8)	Businesses should support inclusive growth and equitable development.	1. R&R Policy 2. NTPC Policy for CSR and Sustainability 3. Initial Community Development (ICD) Policy	Shri D K Patel DIN: 08695490 Designation: Director (HR)
Principle 9 (P9)	Businesses should engage with and provide value to their customers and consumers in a responsible manner.	1. Commercial systems & Procedures	Shri C K Mondol DIN: 08535016 Designation: Director (Commercial)

* Code of Conduct for Board Members & Senior Management Personnel

2. Details of Director/Directors responsible for BR as a whole

a. Details of the Director/Directors responsible for implementation of the BR policy/policies:

S.No.	Particulars	Details
1.	DIN Number	00307037
2.	Name	Gurdeep Singh
3.	Designation	Chairman & Managing Director
4.	Telephone number	011-24360044
5.	E-mail ID	cmd@ntpc.co.in

b. Details of BR head: Same as above

3. Principle wise reply to each question on BR Policy / Policies:

Sl. No.	Question	P1	P2	P3	P4	P5	P6	P7	P8	P9
1.	Do you have a policy/ policies for the Principle?	Y	Y	Y	Y	Y	Y	Y	Y	N
2.	Has the policy being formulated in consultation with the relevant stakeholders?	Y	Y	Y	Y	Y	Y	Y	Y	-
3.	Does the policy conform to any national /international standards?	Y	Y	Y	Y	Y	Y	Y	Y	-

Sl. No.	Question	P1	P2	P3	P4	P5	P6	P7	P8	P9
4.	Has the policy being approved by the Board? If yes, has it been signed by MD/owner/CEO/appropriate Board Director?	Y	Y	Y	Y	Y	Y	Y	Y	-
5.	Does the company have a specified committee of the Board/ Director/ Official to oversee the implementation of the policy?	Y	Y	Y	Y	Y	Y	Y	Y	-
6.	Indicate the link for the policy to be viewed online?	(i)	(i)	(i)	(i)	(i)	(i)	(i)	(i)	-
7.	Has the policy been formally communicated to all relevant internal and external stakeholders?	Y	Y	Y	Y	Y	Y	Y	Y	-
8.	Does the company have in-house structure to implement the policy/ policies?	Y	Y	Y	Y	Y	Y	Y	Y	-
9.	Does the Company have a grievance redressal mechanism related to the policy / policies to address stakeholders' grievances related to the policy / policies?	Y	Y	Y	Y	Y	Y	Y	Y	-
10.	Has the company carried out independent audit / evaluation of the working of this policy by an internal or external agency?	Y	Y	Y	Y	Y	Y	Y	Y	-

(i) Web links for the Policies :

S. No	Policies/Systems	Web link
		Finance
a.	Code of Conduct	http://www.ntpc.co.in/investors/code-of-conduct
b.	Fraud Prevention Policy	http://www.ntpctender.com/about/FraudPolicy.asp
c.	Whistle Blower Policy	https://www.ntpc.co.in/sites/default/files/downloads/WhistleBlowerPolicy2014.pdf
d.	Internal Code of Conduct for prevention of insider trading	http://www.ntpc.co.in/download/internal-code-conduct-prevention-insider-trading-dealing-securities-ntpc-limited
e.	Code of Corporate fair disclosure practices for prevention of insider trading	http://www.ntpc.co.in/download/code-corporate-fair-disclosure-practicces-prevention-insider-trading
f.	Related Party Transaction Policy	http://www.ntpc.co.in/download/related-party-transaction-policy-ntpc
g.	Policy for determination of materiality of events or information for disclosure	http://www.ntpc.co.in/sites/default/files/downloads/NTPC%20%20Policy%20For%20Determination%20of%20Materiality%20of%20events.pdf
h.	Policy on maintenance & preservation of documents	http://www.ntpc.co.in/sites/default/files/downloads/Document%20Preservation%20Policy.pdf
i.	Policy for Determining Material Subsidiaries	http://www.ntpc.co.in/download/policy-determining-material
j.	Dividend Distribution Policy	https://www.ntpc.co.in/sites/default/files/downloads/DividendDistributionPolicyofNTPCLimited.pdf
		Human Resource
k.	CDA Rules	Policy Manual available on NTPC Intranet
l.	Training Policy for Directors of NTPC	https://www.ntpc.co.in/sites/default/files/downloads/training-policy-directors-ntpc.pdf
m.	Human Rights Policy	https://www.ntpc.co.in/sites/default/files/downloads/NTPCHuman%20RightsPolicy_0.pdf
n.	Placement and Transfer Policy	Policy Manual available on NTPC Intranet
o.	R&R Policy	https://www.ntpc.co.in/sites/default/files/downloads/R%26RPolicy2017.pdf

S. No	Policies/Systems	Web link
p.	Initial Community Development Policy	http://www.ntpc.co.in/download/initialcommunity-development-policy-2009
q.	Policy for CSR & Sustainability	https://www.ntpc.co.in/sites/default/files/downloads/NTPC-Policy-for-CSR-2019.pdf
r.	Career development Policy	https://www.ntpc.co.in/en/careers/career-growth-opportunities
	Operations	
s.	Safety Policy	https://www.ntpc.co.in/sites/default/files/downloads/SafetyPolicy2016.pdf
t.	Environment Policy	https://www.ntpc.co.in/sites/default/files/downloads/EnvironmentPolicy2017.pdf
u.	Water Policy	https://www.ntpc.co.in/sites/default/files/downloads/WaterPolicy2017.pdf
v.	Ash Policy	https://www.ntpc.co.in/sites/default/files/downloads/AshPolicy2015.pdf
w.	E-waste Policy	https://www.ntpc.co.in/sites/default/files/downloads/e-wastePolicy2014.pdf
x.	Integrated Plastic Management Policy	https://www.ntpc.co.in/sites/default/files/downloads/Integrated-Plastic-Managment-Policy-2019.pdf
	Technical	
y.	Biodiversity Policy	https://www.ntpc.co.in/sites/default/files/downloads/BiodiversityPolicy2018.pdf
z.	Rain Water Harvesting Policy	https://www.ntpc.co.in/sites/default/files/downloads/NTPC%20Rain%20Water%20Harvesting%20Policy%202018.pdf
	Commercial	
za.	Commercial Systems & Procedures	https://www.ntpc.co.in/sites/default/files/downloads/procurementandworkspolicy.pdf
zb.	Sustainable Supply Chain Policy	https://www.ntpc.co.in/sites/default/files/downloads/Sustainale-Supply-Chain-Policy.pdf

4. If answer against any principle is 'No', please explain why:

Principle 9: All the sub-principles identified under principle -9 are duly followed by your Company through its commercial systems and procedures. However, your Company feels that a separate Policy on Principle -9 is not required because:

- Your Company supplies power to the Bulk Customers (State Electricity Distribution companies) majority of which are owned by the respective State Govt.
- The CERC, while finalizing Tariff and other Regulations engages all Stakeholders and takes views of them. CERC Tariff Regulations and relevant orders are being displayed on CERC Website: www.cercind.gov.in.
- Your Company & our bulk customers i.e. Discoms work under regulated Environment. Your Company strives for supplying cheapest power deploying all resources optimally in best possible ways resulting in well-being of customers & society.
- Your Company being a Government company is also subject to the various checks and balances mechanism such as audits etc.
- CERC while determining the tariff of your Company's stations does prudence check on the costs of your Company.
- Your Company never restricts the freedom of choice and free competition in any manner while supplying bulk Power.
- Needs of the customers is taken into account and accordingly PPA are signed and Allocation of Power is made by Ministry of Power (MoP) as per existing guidelines & Policy. Unallocated quota of power is allocated by MoP as per demand and requirement of different States hence customers are always kept first.
- Central Electricity Regulatory Commission governs power supply rules and regulations, performance and all other commercial parameters related to power generation. Your Company always excels in satisfying customers by adhering to these norms and disclosing all relevant information.

- Issues, if any, regarding operational issues etc. are being discussed and resolved in common forums such as Regional Power Committees.
- Your Company has developed a Customer Satisfaction Index (CSI), which is administered through a questionnaire and based on the feedbacks received, actions are taken.

Your Company regularly engages with the customers and provides value to them in a responsible manner.

5. Governance related to BR

1.	Indicate the frequency with which the Board of Directors, Committee of the Board or CEO to assess the BR performance of the Company.	Within 3-6 months
2.	Does the Company publish a BR or a Sustainability Report? What is the hyperlink for viewing this report? How frequently it is published?	The company publishes both Business Responsibility Report (as a part of annual report) and Sustainability Report annually and can be accessed from the below link. https://www.ntpc.co.in/en/sustainability/compliances-and-reports

Section E: Principle –wise performance

Principle 1: Businesses should conduct and govern themselves with Ethics, Transparency and Accountability

1. Does the policy relating to ethics, bribery and corruption cover only the company? Does it extend to the Group/ Joint Ventures/ Suppliers/ Contractors/ NGOs / Others?

   i. Code of Conduct for Board Members & Senior Management Personnel covers all the Directors and Senior Management Personnel of the Company.

   ii. Fraud Prevention Policy applies to any fraud, or suspected fraud involving employees of your Company as well as representatives of vendors, suppliers, contractors, consultants, service providers or any outside agency(ies) doing any type of business with your Company.

   iii. However, in line with your Company, RGPPL and NTECL, JVs of your Company have also adopted Fraud Prevention Policy and CDA rules.

   iv. CDA Rules are applicable to all employees of your Company and employees posted in JVs/Subsidiaries.

   v. Insider Trading Code is applicable to designated employee of the company.

   vi. Related Party Transaction Policy is framed to ensure the proper approval and reporting of transactions between the Company and its Related Parties.

   vii. The objective of Whistle Blower Policy is to build and strengthen a culture of transparency and trust in the organization and to provide employees with a framework / procedure for responsible and secure reporting of improper activities (whistle blowing) within the company and to protect employees wishing to raise a concern about improper activity/ serious irregularities within the Company.

   viii. The Policy for determination of materiality of events or information for disclosure was framed in terms of Regulation 30 (1) of Securities and Exchange Board of India (Listing Obligations and Disclosure Requirements) Regulations, 2015 (SEBI (LODR).

   ix. The Policy on maintenance & preservation of documents was framed in pursuance to Regulation 9 of the SEBI (LODR).

   x. The Policy for Determining Material Subsidiaries was framed in accordance with the requirement stated under the Listing Agreement.

   xi. The Training Policy aims at providing Orientation & Training programs to be offered to the Board of Directors of your Company.

   To ensure transparency, objectivity and quality of decision making in various operations, the Company has a Vigilance Department headed by Chief Vigilance Officer. The Vigilance set up in the Company consists of Vigilance Executives in Corporate Centre as well as in sites. In sites, the Vigilance Executives report to the Project Head in administrative matters and they report to the Chief Vigilance Officer in functional matters.

Corporate Vigilance Department consists of four Cells as under:

- Vigilance Investigation and Processing Cell
- Departmental Proceedings Cell
- Technical Examination Cell
- MIS Cell

These cells deal with various facets of vigilance mechanism. The vigilance works have been assigned region-wise to Vigilance officers at Corporate Centre (Regional Vigilance Executives) for speedier disposal. Senior officials of Vigilance Department comprising GM (Vigilance), Regional Vigilance Executives and Head of DPC/MIS Cell meet regularly to discuss common issues to ensure uniform working in all Regions. This facilitates transparency, efficiency and effectiveness of Vigilance functionaries by making use of collective knowledge, experience and wisdom of Vigilance Executives as well as breaking of compartmentalization and abridging of strengths & weaknesses.

Fraud Prevention Policy has been implemented in your Company and suspected fraud cases, referred by the Nodal Officers to Vigilance Department are investigated immediately to avoid/ stop fraudulent behaviours as defined in "Fraud Prevention Policy". Whistle Blower Policy has also been in place at your Company as per SEBI guideline to strengthen a culture of transparency and trust in the organization, providing employees with a framework/ procedure for responsible and secure reporting of improper activities (whistle blowing) within the company and to protect employees wishing to raise a concern about improper activity/serious irregularities within the Company. A complaint handling policy is also in place which is designed to provide guidance on the manner in which your Company receives and handles complaints against its employees, suppliers / contractors etc. Integrity Pact has been implemented in your Company since 2009. Presently tenders having estimated value of ₹10 Crore (excluding taxes and duties) and above are covered under the Integrity Pact.

2. **How many stakeholder complaints have been received in the past financial year and what percentage was satisfactorily resolved by the management? If so, provide details thereof, in about 50 words or so.**

During the year 2021-22 total 63 Vigilance complaints were investigated, out of which 52 complaints have been finalized and remaining 11 are under various stages of investigation as on 31.3.2022. Appropriate disciplinary action has also been initiated against the involved employees along with system improvements, wherever found necessary. 414 Preventive Vigilance Checks (Surprise, Routine, CTE Type & Scrutiny of Files) were conducted during the period and recovery of ₹ 99,76,784 was effected against various discrepancies detected during investigation. During the last year a total of 89 Preventive Vigilance Workshops were conducted at various projects/ places of the Company.

Principle 2: Businesses should provide goods and services that are safe and contribute to sustainability throughout their life cycle

1. **List up to 3 of your products or services whose design has incorporated social or environmental concerns, risks and/or opportunities**

i. **Generation of Electricity**: Your Company has been conventionally producing Electricity through Coal and Gas. Many design upgradations such as supercritical and ultra-supercritical based new capacity additions and process innovation such as High Concentration slurry disposal (HCSD), Real time monitoring of Pollutants and Zero Liquid discharge has been done to reduce the environment footprint. With the focus on curbing release of SOx and other undesirable emission into the environment, your Company is undergoing of commissioning of Flue Gas Desulphurisation (FGD) system at various stations. Your Company has also increased its solar, wind and hydro based capacities which has negligible emissions.

ii. **Coal Mining:** Your Company was allocated six coal blocks, namely, Pakri-Barwadih, Chatti-Bariatu & Chatti-Bariatu (South), Kerandari, Dulanga, Talaipalli and Badam by Ministry of Coal. Banhardih coal block was allocated to Patratu Vidyut Utpadan Nigam Ltd. (PVUNL), a JV company incorporated between your Company & Government of Jharkhand.

The estimated geological reserves of the allocated blocks are ~5 billion tonnes with an estimated mining capacity of 71 million tonnes per annum. With these coal blocks, your Company aspires to become one of the largest captive coal mining companies in the country.

Your Company achieved highest ever coal production of 14.02 MMT from Pakri Barwadih, Dulanga and Talaipalli in FY 2021-22, with 27.45 % of growth from FY 2020-21. Also, the highest ever coal dispatch of 14.86 MMT for FY 2021-22, has also seen growth of 35.64% from FY 2020-21. So far, the operational mines have delivered more than 46 Million MT of coal to more than 22 thermal plants of your Company. The coal, from captive coal mines, is much sought after in your Company power stations because of its consistent quality, size and absence of any shale/boulder etc.

Your Company Coal Mining has achieved more than 100% and highest ever Capex utilization of ₹ 1315 crore in FY 2021-22 which is for the first time in Coal Mining. Highest ever profit of ₹ 111 crore was also recorded for FY 2021-22.

Environmental coal mining highlights for year 2021-22

- As a part of green initiative, nursery has been developed in Dulanga mines for a total sapling capacity of 15,000. As on date 6000 saplings have been nurtured. Since inception plantation of 32,400 nos. plants has been carried out.
- Rainwater harvesting has been adopted at two locations i.e. Vocational Training Centre and Coal Laboratory at Dulanga Coal Mining Project which shall conserve approx. 1,000 M^3 of rainwater per year.
- As a part of environmental restoration / protection initiative at Pakri Barwadih Coal Mining Project, 82,569 plants have been planted for FY 2021-22 leading to a total of 2,25,250 plants since inception. Bio-methanation Plant of 100 kg waste/day has been installed under waste to energy initiatives. The biogas produced is being used in the main canteen at Sikri Township.
- Pakri Barwadih Coal Mining Project have installed roof top rainwater harvesting system at office buildings at Langatu site office under water conservation measures. Also, eight nos. of Check dams in the Pakwa and Khora Nala have been constructed for augmenting the ground water of the area.
- Talaipalli Coal Mining Project has planted 21,250 tree saplings. 11 nos. of ground water recharge structures have been constructed at Nayarampur village to recharge ground water to the tune of 73,07,401 M^3 of rainwater per year.

iii. **Green H2/ Alternate Fuel/ Chemical manufacturing:** As part of our business diversification plan, your Company has ventured into alternate fuels and chemicals based on green hydrogen technologies. This would be through integration of existing thermal assets and new investment in clean technologies such as Carbon capture and utilization (CCU), hydrogen production etc. Also green hydrogen technologies like mobility, energy storage etc are also being taken up. Towards that, your Company has planned to take various pilots projects few of which are listed below:

1. Green Hydrogen Mobility
   a. Hydrogen Filling Station at Leh and Delhi for 5 Buses each
   b. Operational 10 Nos. Fuel cell electric buses (FCEV) Buses
2. Green Methanol (10 TPD) at Ramagundam, Telengana
3. Energy storage 50 kW Microgrid at Simhadri (Andhra Pradesh)
4. PNG hydrogen blending at Kawas

2. For each such product, provide the following details in respect of resource use (energy, water, raw material etc.) per unit of product(optional):

a. Reduction during sourcing/production/distribution achieved since the previous year throughout the value chain.

b. Reduction during usage by consumers (energy, water) has been achieved since the previous year.

Raw Material (Energy) Consumption per year:

Energy Source	2019-20		2020-21		2021-22*	
	Quantity	PUC	Quantity	PUC	Quantity	PUC
Coal	169.38 MMT	0.63 kg/kWh	174.96 MMT	0.67 kg/kWh	195.25 MMT	0.67 kg/kWh
Natural Gas	3.16 MMSCMD	0.233 scm/kWh	3.49 MMSCMD	0.224 scm/kWh	1.30 MMSCMD	0.247 scm/kWh
LDO	79,646.20 KL	0.51 ml/kWh (LDO & HFO combined)	84,075 KL	0.49 ml/kWh (LDO & HFO combined)	113991 KL	0.56 ml/kWh (LDO & HFO combined)
HFO	48,895 KL		45,077 KL		48836 KL	
Naphtha	97.9 MT		18,280 MT		8923 MT	
HSD	86.566 KL		284.685 KL		18177.60 KL	

Water Withdrawal per year (in MCM): (** Water calculated on closed loop systems in your Company only)

S. No.	Type of water	Quantity Consumed **		
		2019-20	2020-21	2021-22
1	Total Water withdrawal	559.5	590.89	662.67
2	Per unit withdrawal	3.15 lt/kWh	3.03 lt/kWh	2.96 lt/kWh

Energy saving per year:

	2019-20	2020-21	2021-22
Total Energy saved by the initiatives taken in your Company power plants for energy conservation / efficiency improvement	1622.5 TJ	1643.3 TJ	1693.5 TJ

3. **Does the company have procedures in place for sustainable sourcing (including transportation)? If yes, what percentage of your inputs was sourced sustainably? Also, provide details thereof,**

Coal, Gas, Biomass and water are our key raw materials. We continuously strive to reduce our specific consumption of raw materials in line with our philosophy of sustainable production and consumption. Optimization of raw material consumption translates into value for our shareholders, as well as for our communities. This enable us to also reduce the tariff of electricity making it affordable for all and also reduce the burden on local resources by minimizing extraction and waste generation.

The Following procedures are in place for the sustainable sourcing of coal by your Company for its different power station:

Long Term Fuel Supply Agreements

- In line with the Model Coal Supply Agreements agreed between the Company and Coal India Ltd. (CIL) in 2009, 2012 and under Shakti Policy, Long-term Fuel Supply Agreements (FSAs) are in place with the subsidiary coal companies of CIL for an Annual Contracted Quantity (ACQ) of 167.85 Million Metric Tonnes (MMT), as on 31.3.2022 for the existing thermal stations. In addition, FSAs are in place with Singareni Colliery Company Ltd. (SCCL) for Ramagundam and Solapur Unit-2 for an ACQ of 13.74 MMT. FSAs are generally valid for a period of 20 years with a provision of review after every 5 years.
- In FY 2021-22, your Company has signed Long Term FSA for 2.54 MMT per annum with SCCL for Solapur Unit-2 (660 MW) pursuant to its coal linkage transfer from distantly located Mahanadi Coal fields Limited (MCL) sources, thereby reducing cost of generation from the plant. Further, as per decision of SLC(LT) to enhance ACQ of eligible plants upto 100% of normative quantity (i.e., quantity for 85% PLF), your Company has signed supplementary/amendment agreements with coal companies for various plants viz., Rihand-III Unit-6, Vindhyachal-IV Unit-11&12, Mouda-II Unit-3&4, Unchahar-IV Unit-6, Gadarwara Unit-1&2 for total additional quantity of 1.75 MMT per annum.
- Due to suspension of coal mining activities at NEC (CIL) and pursuant to the Govt. policy of linkage rationalization, coal linkage of Farakka-I&II Unit-1 to 5 with NEC has been transferred and FSA is signed with ECL, whereas for Bongaigaon Unit-1&2, the FSA with NEC has been transferred and signed with CCL (1.171 MMT) and ECL (0.15 MMT).

Fresh long-term coal linkages under para B(i) of SHAKTI policy

- In FY 2021-22, based on your Company request, SLC (LT) dated: 22.12.2021 recommended coal linkage for Khargone Unit-1 (660 MW) from CIL. Subsequently, CIL has allocated linkage for annual quantity of 3.336 MMT from MCL. Issuance of Letter of Assurance from MCL and signing of FSA is expected in FY 2022-23.
- Pursuant to termination of existing Letter of Assurances (LOA) of Barh-I (Unit 2 & 3) and North Karanpura (Unit 1 to 3) as per the terms of para A(i) of SHAKTI policy, Standing Linkage Committee (Long-term), SLC(LT) in its meeting held on 29.3.2022 has recommended for grant of fresh coal linkages for the plants from CIL under para B(i) of SHAKTI policy.

Coal under Bridge Linkage and other short-term Memorandum of Understanding (MoU)

- Bridge linkages of Barh-II, Tanda-II, Lara, Darlipali, Kudgi and Barauni-II: In terms of the Govt. policy of Bridge Linkage, your Company's plants viz. Barh-II, Tanda-II, Lara, Darlipali, Kudgi and Barauni-II were allocated Bridge Linkages to bridge the gap between coal from linked captive mine and requirement of the plant. The bridge linkages were granted by SLC(LT), MoC for Tanda-II, Lara and Darlipali each upto 2022; Barh-II upto 2023 and Barauni-II upto 1.9.2022. Based on above and coal quantification by CCO as per tapering based on approved mining plan of respective linked mines, MoUs with allocated coal companies are valid upto March'2022.

In view of expiry of Bridge Linkages, your Company has taken up before Ministry of Power / Ministry of Coal for further extension of Bridge Linkages considering present production plan of the linked captive mines. Further, Ministry of Power has recommended to Ministry of Coal for grant of extensions of the Bridge Linkages. The matter was discussed in the SLC(LT) meeting held on 29.3.2022. However, it was deliberated to consider the matter in next SLC(LT) after examination of your Company proposal by Nominated Authority, Ministry of Coal.

In relation to Kudgi, against your Company request for grant of long-term coal linkage, SLC(LT) in the meeting held on 29.3.2022 has recommended for extension of Bridge Linkage by another 01 year (w.e.f. April'2022) or till surrendering of the linked mine, Bhalumuda is completed.

- Tapering linkage of Telangana Ph-I (2X800 MW): Pursuant to transfer of tapering linkage of Telangana Ph-I (granted against linked captive mine, Mandakini-B) from WCL cost-plus to SCCL, your Company in the FY 2021-22 has signed MoU for 6.846 MMT per annum with SCCL, with validity upto March'2023.
- MoU with SCCL: A bilateral MoU was signed with SCCL for FY 2021-22 for a quantity of 8.00 MMT on best effort basis for supply of coal to Kudgi plant under Bridge Linkage and other your Company stations (except Ramagundam) as per requirement.
- MoU with NLC India Ltd.: Pursuant to Govt. notification of Mines and Minerals (Development and Regulation) (Amend) Act, 2021 and Mine Concession (Amend) Rules, 2021, permitting commercial sale of captive mine coal up to a percent of the annual production after meeting the end use requirement, your Company has entered into a bilateral MoU with NLC India Ltd. for supply of coal from its Talabira II & III mine to your Company plants during period Oct-Dec'2021. Further, your Company and NLC India Ltd. have to entered into bilateral MoU for supply of Talabira coal for 3 years for annual quantity of 3 MMT on best effort basis.

Agreements for supply of imported Coal

- As per the directions of MoP & CEA, in order to mitigate the shortage of domestic coal by blending with Imported coal, package quantity of running contracts was enhanced by 1.45 MMT during 2021-22.
- As per directions of MoP, your Company is procuring 16 MMT of Imported coal in 03 tranches of 7.0 MMT, 5.0 MMT and 4.0 MMT. Contracts have already been awarded for procurement of 6.75 MMT while procurement of 6.25 MMT is under process. For JVs, contracts have been awarded for 1.8 MMT while 0.67 MMT is under process.

Domestic Coal and Imported Coal

During 2021-22, your Company Received 224.14 MMT of Coal as against 191.26 MMT in 2020-21. Out of 224.14 MMT of Coal, 198.54 MMT was from Annual contracted quantity of Coal, 8.43 MMT Through Bridge Linkage/SCCL Bilateral MoU, 14.70 MMT from Captive Mines and 2.47 MMT from Import Coal.

Biomass Co-Firing Utilization of Agro residue for power generation & reduce pollution:

As part of its commitment towards clean environment, your Company has taken a new initiative to utilise agro residue for power generation. This is intended to cut down carbon emissions and also to discourage crop residue burning by farmers after harvesting by adding economic value to the crop residue and providing extra income to farmers and employment in rural sector. Biomass co-firing is a unique method to utilize coal based power plant infrastructure to produce renewable energy by simply replacing some of the coal with biomass based fuel. Being carbon neutral fuel, biomass co-firing is a technology recognized by UNFCCC as a measure of reducing greenhouse gas emission.

After successfully demonstrating biomass co-firing at your Company Dadri plant, your Company has started commercial scale biomass co-firing at other your Company stations also. Your Company has forayed into large scale procurement of biomass for Co-firing by issuing a Long Term (LT) tender on 12.1.2022 for procurement of ~9.2 MMT biomass for 04 projects of your Company over the period of seven years. It is expected to place the award under the long term tender before the next harvesting season in October-November.

Besides LT tender, Your Company has awarded contracts for procurement of 4.56 LMT of biomass pellet under short term contracts (9 months). Supply has already started at 7 your Company stations under short term contracts and also under orders placed for 280 TPD for four years (approx. 4 lakh tonnes) at Dadri plant. The Company has already fired ~77,450 metric tonnes of biomass pellets at its various plants by 20 July'22.

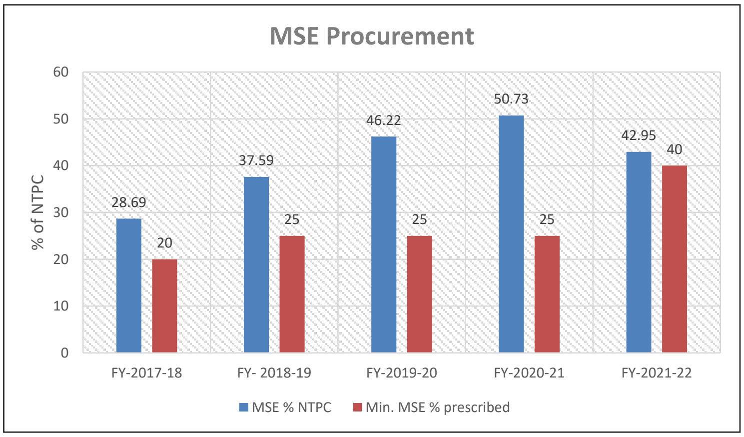
Significant achievements of 2021-22

- Total Domestic Coal Receipt (COD Stations) in FY 2021-22 was 221.67 MMT.

- 23.06 MMT of coal diverted to critical stations during FY 21-22 under policy of flexible utilization of domestic coal to address AFC under recovery and generation loss.
- Weighment of 100% incoming rakes achieved.
- Transit loss for supply through Rail in FY 21-22 - 0.4%.
- Timely tie up of RLNG resulted in 'Nil' DC loss at Gas stations due to lack of fuel availability. Annual Fixed cost under recovery of Gas stations is NIL for FY 21-22.
- Long term RLNG consumed in year 2021-22 was 218.6 MMSCM.

4. Has the company taken any steps to procure goods and services from local & small producers, including communities surrounding their place of work? If yes, what steps have been taken to improve their capacity and capability of local and small vendors?

The Company adopts fair, equitable and transparent tendering procedures. To encourage Indian bidders and suppliers, provisions regarding price preference and deemed export benefits are stipulated in the bidding documents as per the extant policy of Government of India. Company has taken various initiatives to encourage participation in tender for local & small manufacturer including SMEs. Your Company Limited is a fully compliant CPSU on MSME SAMBANDH portal.



As per Public Procurement Policy for MSEs Amendment Order dated 9th November 2018, Every Central Ministry /Department / PSUs shall set an annual target for 25% procurement from MSE Sector. Out of the total annual procurement from Micro and Small Enterprises, 3 per cent from within the 25 per cent target shall be earmarked for procurement from Micro and Small Enterprises owned by women. As per DC (MSME) letter dated 31st August 2021, the target for your Company is revised to 40% minimum procurement from MSE vendors against the exemptions.

Some highlights are as below:

- Your Company exceeded the target of 40% procurement from MSEs in FY 2021-22 as per Public Procurement Policy (PPP) 2012.
- The MSE vendor count configured in SAP is 7,627 till March 2022 end.
- Your Company has dedicated portal for MSE vendors for collating vendor database for items reserved exclusively for procurements from MSEs. The portal mitigates the long pending requirement of your Company Units for a validated vendor base for reserved items.
- Your Company is on-board to all 3 Trade Receivable electronic Discounting System (TReDS) i.e. M1Xchange, INVOICEMART and RXIL. Trade Receivables worth ₹ 0.61 crore Discounted/paid through TReDS till March'2022 end.

5. Does the company have a mechanism to recycle products and waste? If yes what is the percentage of recycling of products and waste (separately as <5%, 5-10%, >10%). Also, provide details thereof, in about 50 words or so.

Our Product viz. Electricity gets completely consumed and hence there is no scope of its recycling. Guidelines have been issued in line with National Environment Policy for disposal of hazardous wastes from your Company power stations. The hazardous wastes generated at our power stations such as used transformer oil, used lubricants, lead acid batteries etc. are sold only to government approved recyclers or given back to the sellers for recycling under buy back arrangements, which takes care of 100% recycling of such wastes.

Sustainable ash utilization is one of the thrust areas of your Company's activities and the Company strives to maximize the ash utilization at all your Company's power plants. Your Company has introduced Ash Policy, which is a vision document dealing with the ash utilization issue in an integral way from generation to end product. This policy aims at maximizing utilization of ash for productive usage along with fulfilling social and environmental obligations, as a green initiative in protecting the nature and giving a better environment to future generations.

During FY-2021-22, approximately 83.19 million tons of ash was generated by your Company Group stations and 81.67% i.e. 67.94 million tons of ash had been gainfully utilized in various areas such as issue to industries for cement, concrete, asbestos products, bricks/ blocks making, road embankment construction, ash dyke raising, mine filling, land development works etc. Ten stations have achieved 100% ash utilization.

Principle 3: Businesses should promote the wellbeing of all employees

1. **Number of Employees:**

Type of employee	Employee category	Male	Female
Permanent employee	Board of Directors and KMP	6	0
	Top/ Senior management	34	1
	Middle management	3672	181
	Lower/Junior management	7049	589
Total permanent Employee		**10761**	**771**
Permanent workers	Non-Executive	5509	450
Other than permanent employee	(Other than permanent such as interns, trainees / apprentices, part time employees/Fix term employee, etc)	801	107

2. **Number of Employees hired on Temporary / Contractual / Casual basis:**

   Your Company hires employees on FTEs. Its included in above table in the "Other than permanent employee"

3. **Number of permanent Women Employees :** 1221

4. **Number of permanent Employees with Disabilities :** 492

5. **Do you have an employee association that is recognized by management:**

   Your Company is a multi-unit organization. Association(s) comprising of executives of your Company need not be recognized in the absence of any statutory mandate. Workmen of various your Company Units have formed unions. The same are accorded recognition by your Company as per applicable law / practice.

6. **What percentage of your permanent employees is members of this recognized employee association?**

   About 70% of the permanent employees in workmen category are members of the recognized union of workmen.

7. **Please indicate the Number of complaints relating to child labour, forced labour, involuntary labour, sexual harassment in the last financial year and pending, as on the end of the financial year.**

No.	Category	No. of previous year pending complaints	No. of complaints filed during the	2021-22
1	Child labour / forced labour / involuntary labour	NIL	NIL	NIL
2	Sexual harassment	0	2	0
3	Discriminatory employment	NIL	NIL	NIL

8. **What percentage of your under mentioned employees were given safety & skill up-gradation training in the last year? (Excluding joint venture companies)**

   Safety Training

Stations	Type	No. of Prog.	No. of persons trained in safety
Your Company	Employees	383	7644
	Workers	5041	172726

Skill Upgradation Training

Categorisation	Employee Categories	% of employees who attended skill development trainings
Permanent employee	Board of Directors and KMP	0%
	Top/ Senior management	71%
	Middle management	88%
	Lower/Junior management	79%
Permanent workers	Workmen (Non-Executive)	82%
Other than permanent employee	(Other than permanent such as interns, trainees / apprentic-es, part time employees/Fix term employee, etc)	39%

Principle 4: Businesses should respect the interests of, and be responsive towards all stakeholders, especially those who are disadvantaged, vulnerable and marginalized.

1. **Has the company mapped its internal and external stakeholders?** Yes

2. **Out of the above, has the company identified the disadvantaged, vulnerable & marginalized stakeholders**

   The Company has a well structure R&R and CSR Policy to take care of the disadvantaged, vulnerable and marginalized people in and around the plants.

   Vulnerable category of persons has been detailed in your Company's CSR and R&R Policies. These include old-aged, women, PCPs, SC/ST/OBC etc. They are identified through Socio Economic Survey (SES), Need Assessment Survey (NAS) and other consultations with the stakeholders etc. at the time of formulation of CSR/ R&R Plans.

3. **Are there any special initiatives taken by the company to engage with the disadvantaged, vulnerable and marginalized stakeholders? If so, provide details thereof, in about 50 words or so.**

   Yes,

   Your Company has always been sensitive to the needs of disadvantaged, vulnerable and marginalized stakeholders. As part of R&R, in addition to the entitlements and packages as envisaged for PAP's, your Company makes special efforts for the welfare measures for this section of the society in the neighborhood community of your Company locations. The measures include special vocational training programmes for making them self-employable, priority in engagement for suitable jobs, pension under Widow Pension Scheme, old age pension scheme etc as per provisions outlined in extant policies or as decided in Village Development Advisory Committee VDAC/ R&R plan. Furthering these efforts under CSR, your Company regularly undertakes Skill up gradation programs for improving employability of youth/ women in the neighborhood of its operations, some of which also fall in the Aspirational Districts identified by 'NITI Aayog'. Your Company has also taken up the activities for women empowerment, construction of SC/ST multipurpose halls & hostels, relief through distribution of various articles & support to orphanages & old age homes in the vicinity of its stations. Some of the initiatives to address the needs of differently abled include Information and Communication Technology (ICT) Centres and Disability Rehabilitation Centres by NTPC Foundation and inclusive education, distribution of equipments like tricycles, wheelchairs, hearing aids, vocational training, scholarship etc.

Principle 5: Businesses should respect and promote human rights

1. **Does the policy of the company on human rights cover only the company or extend to the Group / Joint Ventures / Suppliers / Contractors / NGOs / Others?**

   The Human Rights policy of your Company is applicable to all its employees posted in various stations, projects, offices, JVs and Subsidiaries. Human Rights provisions are also built in our bidding documents for supply cum erection and civil packages invited on competitive bidding basis covering our suppliers and contractors.

2. **How many stakeholder complaints have been received in the past financial year and what percent was satisfactorily resolved by the management?**

   No complaint on human rights, such as child labour, forced labour, involuntary labour, sexual harassment, discrimination, rights of the disabled etc. was pending as on 31.3 2022. (Refer principle 3)

Principle 6: Business should respect, protect, and make efforts to restore the environment

1. **Does the policy relate to Principle 6 cover only the company or extends to the Group / Joint Ventures / Suppliers / Contractors / NGOs /Others?**

   All the policies of your Company as mentioned under principle 6 and their implementation thereof covers all NTPC business units as well as its JVs and subsidiaries. However, the Suppliers / Contractors / Other stakeholders are free to adopt the same voluntarily.

2. **Does the company have strategies/ initiatives to address global environmental issues such as climate change, global warming, etc? If yes, please give hyperlink for webpage etc.**

   Yes,

   Your Company is one of the premier PSU under MoP and has been doing its bit in tackling the issue of global warming & climate change. Your Company, as a responsible global citizen, has taken various steps i.e readjustment of NTPC's fuel / power generation mix by adopting more renewables, introduction of clean coal technologies etc. in line with various GoI missions under National Action Plan on Climate Change (NAPCC).

   Your Company low carbon initiatives can be categorized into following broad categories:

   i. Re-adjustment of NTPC's fuel/ power generation mix

   ii. Introduction of Thermodynamically Efficient Technologies

   iii. Renovation & Modernization of old power stations

   iv. Induction of advanced clean coal technologies for power generation

   v. Establishment of NETRA for addressing climate change concerns

3. **Does the company identify and assess potential environmental risks?**

   Your Company has an established system to identify & assess the potential environmental risk at stations by MRM (Management Review Meeting) at Station, Region, Corporate Centre by RMC (Risk management committee) and Management Committee Meeting (MCM). RMC consist member of Director's level, and review meeting is organised on quarterly basis to discuss, elaborate and mitigate the issues, which may cause potential environmental risks. Further an apex meeting (Management committee) is chaired by CMD for solving the critical issues and ensure compliance without deviation. Your Company risks are mainly 100% ash utilisation, implementation of new norms within time bound manner and Court cases related to Environment in NGT and higher courts. All Environmental related risks are regularly identified, assessed, deliberated at appropriate level and action plans are prepared and implemented in time to mitigate such risks, if any.

4. **Does the company have any project related to Clean Development Mechanism? If so, provide details thereof, in about 50 words or so. Also, if yes, whether any environmental compliance report is filed?**

   Your Company is pioneer in undertaking climate change issues proactively. The company has taken several initiatives in CDM Projects in Power Sector.

   Five of its renewable energy projects viz. 5 MW Solar PV Power Project at Dadri, 5 MW Solar PV Power Project at Port Blair (A&N), 5 MW Solar PV Power Project at Faridabad and 8 MW small hydro power project at Singrauli and 50 MW Solar PV Plant at Rajgarh (MP) have already been registered with United Nations Frame Work Convention on Climate Change (UNFCCC) CDM Executive Board.

   15 MW Solar PV Power project at Singrauli, 10 MW Solar PV project at Talcher and 10 MW Solar PV Power Project at Unchahar is registered in UNFCCC CDM Programme of Activities (PoA).

   Registration work is under progress for 200 MW capacity of 250 MW Anantpur solar Power project under GCC Program (voluntary mechanism promoted by the Gulf Organization for Research and Development which has been introduced recently) and 1,502,416 ACCs are expected to generated after verification exercise for first 05 years.

   Further, Registration work of 22 nos. (3202 MW) RE project(s) is proposed to be taken up under GCC (Global Carbon Council) program.

   6173 nos of Certified Emission Reductions (CERs) for 5 MW Solar PV Power Project at Port Blair (A&N) has been issued by UNFCCC CDM Executive Board. Another 5842 nos. of CERs have also been issued by UNFCCC CDM Executive Board for 5 MW Solar PV Power Project at Dadri and 21011 nos. of Certified Emission Reductions (CERs) has been issued for 5 MW Solar PV

Power Project at Faridabad. Further, 80278 nos. of CERs are expected shortly, after completion of verification exercise of 50 MW Rajgarh Solar project.

Further, Registration of 50 MW Solar PV power project at Anantpur, 260 MW Solar PV power project at Bhadla, 250 MW Solar PV power project at Mandsaur and 50 MW Wind power project at Rojmal has been done in Verified Carbon Standard (VCS) program. A total of 1085005 and 2089638 Voluntary Emission Reduction (VERs) has been issued for these projects after completion of 1st and 2nd Verification exercise. Further, 1176114 VERs are expected shortly after completion of 3rd verification exercise for these projects.

Prior consideration forms have been sent to UNFCCC and MOEF for our upcoming following Solar Projects : 140 MW and 85 MW at Bilhaur, 20 MW at Auraiya,100 MW floating solar at Ramagundam, 25 MW floating solar at Simhadri, 70 MW & 22 MW floating solar at Kayamkulam, 160 MW at Jetsar, 20 MW floating solar at Auraiya, 20 MW at Rihand, 1692 MW CPSU Scheme (Trench-I & Tranche-II projects), 90 MW at Anta and 23 MW at Solapur.

Further, Prior consideration forms have also been sent for NREL solar projects (Wholly owned subsidiary of NTPC Limited): 150 MW at Chhatagarh, 320 MW at Bhensara, 150MW at Limdi, 200MW at Waghodia, 150MW wind energy project at Dayapar, 90 MW at Anta, 23 MW at Solapur and 735 (3x245MW) Nokh.

5. Has the company undertaken any other initiatives on – clean technology, energy efficiency, renewable energy, etc? Y/N. If yes, please give hyperlink for web page or write up.

Yes, the company has taken up several initiatives for clean technology, energy efficiency and renewable energy. Details are as follows:

R&D Activities:

Your Company, fully recognize R&D as the cornerstone of technological advancement. Therefore, R&D has been incorporated in the long-term vision and strategy for the benefit of the company and society. Your Company has been assigning more than 2% of PBT consistently for R&D related activities.

NETRA collaborates with leading institutes, technology players and service providers both at national and international level. A Research Advisory Council (RAC) comprising of eminent scientists and experts from India and abroad has been constituted to steer NETRA for high end research. In-house Scientific Advisory Council (SAC) has also been constituted to provide directions for improving plant performance & reducing cost of generation.

The focus areas of NETRA are - Efficiency Improvement & Cost Reduction; New & Renewable Energy; Climate Change & Environmental protection which includes water conservation, Ash utilization, Carbon capture and utilization & Waste Management. NETRA also provides Advanced Scientific Services to its stations and other utilities in the area Non-Destructive Examination (NDE), Metallurgy & Failure analysis, Oil/water chemistry, Environment, Electrical, Computational Fluid Dynamics(CFD), etc. for efficient and reliable performances. NETRA laboratories are ISO 17025 accredited and NETRA NDT laboratory is also recognized as Remnant Life Assessment Organization under the Boiler Board Regulations, 1950.

NETRA has taken up major projects for addressing concerns such as environment and climate change, reliability and efficiency improvement, new and renewable energy etc.

NETRA has taken projects for reducing carbon footprints by means of carbon capture and utilization. 10 TPD CO_2 to Methanol project is aimed at capturing CO2 from existing thermal plant and its utilization for production of valuable product such as Methanol. Work is going on for the indigenization of catalyst and reactor for conversion of CO_2 to Methanol plant.

R&D, lab-scale and pilot-scale demonstration projects such as development of Sea water Electrolyser, High Temperature steam electrolysers, Static hydrogen storage and Compression etc have been taken up for hydrogen technology.

NETRA is working on development of innovative water technologies such as Non Thermal Forward Osmosis, Electro-Dialysis Reversal, Desalination etc for the processing of STP and Sea water.

Urban waste is a major problem in India. MSF-RDF gasification plant shall enable MSW and Agri-waste utilization and can help in managing waste better.

Continuous work is going on for the development of new products and processes for bulk Ash utilization and as a result novel technologies & products such as Geo Polymeric (GPC) Road, Geo polymeric Pavers & Tetrapod etc have been successfully developed and implemented.

Through advance scientific services like Non-Destructive Evaluation, CFD analysis and Metallurgy & Failure Analysis, etc., NETRA is working constantly for improvement in reliability, efficiency and as well as for reduction in O&M cost by the diagnosis of faults. NETRA is also working on cutting-edge technologies such as development of robotics solutions for inspection of CW Duct & water pipeline, visual inspection of boiler headers, boiler 1st pass internal inspection. Asset Digitization & Monitoring through Drones, development of novel sensors to increase reliability, safety and to reduce inspection & monitoring time as well as manpower requirements.

NETRA activities have helped the stations in the analysis of failures and their prevention. Techniques developed by NETRA are implemented at stations wherever required. Regeneration treatments of resins, chemical cleaning/treatment and corrosion control measures supported the stations in improving the efficiency and availability of boilers and various heat exchangers/ cooling towers, etc.

Total R&D expenditure during FY 2021-22 is ₹ 222.48 crore.

Renewable Energy:

Your Company is taking various steps to make its energy portfolio greener by adding significant capacities of Renewable Energy(RE) Sources. By 2032, the company plans to have a minimum of 60 GW capacity through RE sources constituting nearly 45% of its overall power generation capacity.

Description	NTPC Group EPC Mode's Own Projects (10142 MW)			Outsourced Projects (6743 MW)
	NTPC	NTPC REL	THDC & NEEPCO	NTPC
Commissioned	1660	--	192	4783
Under Execution	2470	970	--	790
Under Tendering	300	3295	--	1170
Capacity won	--	1255	--	--
Total	**4430**	**5520**	**192**	**6743**

6. **Are the Emissions / Waste generated by the company within the permissible limits given by CPCB/SPCB for the financial year being reported?**

   Your Company is committed for compliance emission, effluent parameters and waste management and upgradation of technology is in regular practice as per directions received from regulatory agencies. All emission and effluent parameters are being monitored through online system by regulatory agency. These parameters are being maintained well within the permissible limits during the FY 2021-22.

7. **Number of show cause/ legal notices received from CPCB/ SPCB which are pending (i.e. not resolved to satisfaction) as on end of Financial Year.**

   There is no pendency of any case of show cause /notices & complaints received from CPCB/SPCB at the end of the financial year 2021-22.

Principle 7: Businesses, when engaged in influencing public and regulatory policy, should do so in a responsible manner

1. **Is your Company a member of any trade and chamber or association? If Yes, Name only those major ones that your business deals with.**

   Yes.

   The detail of major corporate membership with which Company had a business dealing are as under:

   SCOPE, FICCI, CII, TERI, WEC, CBIP, IERE, POWER SECTOR SKILL COUNCIL, EPRI, ICSI, POWER HR FORUM, ICC, IFC etc.

2. **Have you advocated / lobbied through above associations for the advancement or improvement of public good? if yes specify the broad areas (drop box: Governance and Administration, Economic Reforms, Inclusive Development Policies, Energy security, Water, Food Security, Sustainable Business Principles, Others).**

   Your Company is a member of World Energy Council. Chairman and Managing Director of your Company is also an ex-officio Member Secretary WEC India and its international counterpart WEC work towards sustainable use and supply of energy. Their work enables promoting policies which balance Energy Security, Energy Equity (Energy access/inclusive growth) and Environmental Sustainability.

Principle 8: Businesses should support inclusive growth and equitable development

1. **Does the company have specified programmes / initiatives / projects in pursuit of the policy related to Principle 8? If yes details thereof.**

   Yes

   To achieve the aim of inclusive growth and equitable development, the Company has a comprehensive Resettlement & Rehabilitation (R&R) policy covering community development (CD) activities, which has been revised and updated from time to time. CD activities in green field area are initiated as soon as project is conceived and thereafter extensive community / peripheral development activities are taken up along with the project development. Separate CSR & Community Development Policy, formulated in July 2004 and Sustainability Policy formulated in Nov 2012 have been combined in 2015 and revised in 2016 as "NTPC Policy for CSR & Sustainability" in line with Companies Act, 2013 and DPE Guidelines for CSR. It covers a wide range of activities for inclusive growth including implementation of key programmes through NTPC Foundation. Focus areas of NTPC CSR & Sustainability activities are Health, Sanitation, Drinking Water, Education, Capacity Building, Women Empowerment, Social Infrastructure Development, support to Physically Challenged Person (PCPs) and activities contributing towards Environment Sustainability.

2. **Are the programmes/projects undertaken through in-house team/own foundation/external NGO/government structures/ any other organization?**

   Programmes are undertaken by well-defined in - house team through specialized agencies, NGO's, government agencies/ bodies etc. Some of the activities are carried by NTPC Foundation.

3. **Have you done any impact assessment of your initiative?**

   Yes

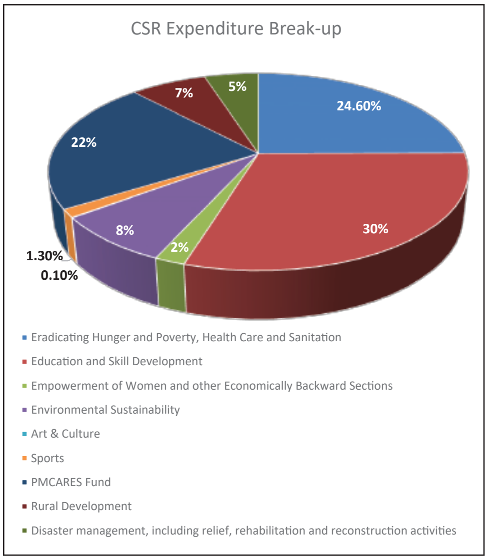
   Effectiveness of CSR & Sustainability programmes are assessed through both internal & external impact evaluations. Internal audits are regularly carried out to verify effectiveness of implementation. Social Impact Evaluation (SIE) is done through credible external agencies for gauging the impact of CSR & Sustainability initiatives. Findings of SIE form the basis for initiating improvement actions and formulating future schemes/plans. Every two-years, your Company conducts a third-party Social Impact assessment for all its major CSR activities at various operating stations. During the Year 2021-22, your Company has taken up Impact Evaluation at 12 locations where periodicity was due. your Company is flexible & open approach in this regard coupled with community participation and with local administration & village Panchayats leads to successful adoption and acceptability of initiatives by community.

4. **What is your Company's direct contribution to community development projects- Amount in INR and the details of the projects undertaken?**

   Yes

   Company has specified programmes for inclusive growth & equitable development not only at station level but also at National level. Details of the programmes and spending are listed below:

   Total spending on CSR – During the FY 2021-22 your Company has spent an amount of ₹ 356.72 crore against the mandated CSR expenditure of ₹ 281.80 crore. Out of the said CSR Expenditure, your Company, in alignment with the guidelines issued by DPE on common Annual theme for CPSEs of "Health & Nutrition" for FY 2021-22, has spent an amount of ₹ 198.95 Crore i.e. 70.60% which is more than the prescribed 60% on Annual Theme.



I Education

- Girl Empowerment Mission (GEM) – A four-week residential workshop for girl children in the age group of 10 to 12 years. The entire cost of the program including their boarding & lodging in a safe & secure environment of these girl children is borne by your Company. Further, free education of around 180 girl students admitted to different your Company Township Schools. These students were selected at Four-weeks residential workshop under Girl Empowerment Mission (GEM) flagship programme of your Company held during summer vacation of 2019 at 23 your Company projects/ stations wherein more than 1800 girls participated.
- In the year 2022, the said programme is being conducted at 33 your Company Projects / stations/ JV & Subsidiaries where around 2400 girls are participating.

- Utkarsh Scholarship Scheme for students:
  - who have passed Xth and XIIth Standard from Govt./ Govt. aided Schools in target villages of your Company Projects/ Stations including schools in your Company townships.
  - Pursuing full time studies in Govt. ITIs in district(s) where your Company Project/ Station are located and ITIs adopted/ set up by your Company.
  - Pursuing full time BE/ B. Tech in IITs, NITs and Govt. Engineering Colleges.
  - Pursuing full time MBBS in Govt. Medical colleges.
- Your Company has established Smart Classes and has taken many initiatives in various Government schools located in the vicinity of its Plants and Stations for improving the learning level of the school children.
- Your Company Sipat project has so far sponsored education to more than 300 Baiga Tribal students for getting admission into engineering and medical colleges through Commissioner, Tribal Welfare, Chhattisgarh.
- Your Company Korba project (aspirational district) is supporting the holistic education to 30 Special Backward Tribe PahadiKorwa students every year from class VI to XII.
- Your Company is supporting the construction of facilities for tribal school in Chapki, Distt. Sonebhadra (UP).
- Your Company is supporting development of school infrastructure in the existing school at Govindnagar, PaliyaPipariya, Block Bankhedi, Hoshangabad, MP.
- Your Company has set up Library (Pustakalaya Project) in 10 Primary Schools of Arunachal Pradesh and Infrastructure augmentation by providing benches/desks in 50 Govt. Schools at Supaul, Bihar.
- Your Company has set up an English Medium School through Shree Ramakrishna Ashram, M. Rampur, Kalahandi.
- Your Company has extended its support for Development, Renovation, and Advancement project of GHSS Munderi, District Kannur, Kerala.
- Your Company has extended its support in providing class-room desks in Govt./ Aided schools in Dharwad Parliamentary area, Dharwad, Karnataka.
- Supporting education in rural areas by augmenting and strengthening school infrastructure including additional classrooms, science labs, libraries, kitchen sheds, providing assets like furniture, computers, separate toilets for girls and boys, water coolers, filters etc. Your Company also ensures the right to education of children from the underprivileged sections of society by providing them with scholarships, study material & uniforms etc.
- 34 Schools in your Company Townships provide subsidized education primarily to rural community children benefitting about 20,000 students of neighbourhood areas.

II Health

COVID – 19 Relief Measures:

- A financial support of ₹250 Crore in FY 2020-21 and ₹80 Crore in FY 2021-22 to Prime Minister's Citizen Assistance and Relief in Emergency Situations Fund (PMCARES) has been made for Covid-19 and other Services/ Activities.
- Established Covid Care Centre (with oxygen) at Badarpur (50 beds) and PMI, Noida (100 beds) and Dadri (30 beds).
- Installed PSA type Oxygen generation plants at various locations like Distt. Hospital Raebareli, Distt. Hospital Robertsganj, Distt. Hospital Sonebhadra, CHC Chhabra, Narsingpur and Ambedkarnagar.
- Support to DPE for procurement of items for COVID-19 Vaccination Programme for the State of Bihar and Uttar Pradesh (UP).
- COVID Care facilitates established with required oxygen support at various your Company Hospitals at Stations/ Projects.
- Construction of Burn Unit at AIIMS Patna, AIIMS Bhubaneshwar and King George Hospital, Lucknow.
- Construction of Medical College & Hospital (400 beds with prov. of upgradation to 500 beds) at District Sundergarh, Odisha under Community Development Plan.
- Support for 5 ambulances to Govt. of Uttarakhand and 4 ambulances to District Administration, Patna.

- Support for Ultraviolet based sanitization technology for PPE kits being developed by IIT Delhi & Chakr Innovation.
- Provided 663 Infrared Thermometers to ASHA workers and 200 KVA DG set provided to District COVID Hospital at Gopeshwar, Uttarakhand.
- Eight (08) Ventilators installed at Hazaribagh Govt. Hospital.
- Support to Collector Khargone for 15 COVID ICU beds at District Hospital Khargone & 250 Oxygen Bed Set-ups in 5 Blocks of Khargone.
- Support to District Administration, Chatra for PPE Kits, Sanitizers and Surgical Masks.
- Provided 25 nos. Oxygen Concentrators to district hospital Barmer.
- Proactive relief measures such as regular sanitization of villages, distribution of PPE kits to health professionals, distribution of face masks to villagers, and supply of groceries to the community.

Promotive, Preventive, & Curative Healthcare:

- Your Company hospitals has provided quality primary, secondary and referral care along with diagnostic services to more than 3,00,000 patients from the community every year.
- Health related infrastructure support is being extended to various PHCs, CHCs and District Hospitals located in the vicinity of your Company Projects/Stations.
- Your Company supports the local administration by regularly carrying the works of fogging, spraying etc. in villages located in its vicinity and also distributes mosquito nets for prevention of vector borne diseases.
- Your Company makes primary care accessible, affordable and inclusive through health camps – both general and specialized (Eye, Respiratory Diseases/ Cancer detection etc.), outreach services through mobile health clinics and by augmenting and strengthening rural health infrastructure. About 3000 surgeries are performed every year in these camps.
- Your Company is supporting the availability of adequate tertiary care in the country:
  - Support for Development of King George Hospital, Visakhapatnam.
  - Support for construction and equipment for 3rd Floor and diagnostic labs at National Cancer Institute Nagpur, Maharashtra.
  - Support to L V Prasad Institute for construction of operating room complex at MTC Campus Bhubaneshwar.
  - Setting up of Special burn unit at KGMU Lucknow, AIIMS Bhubaneshwar and AIIMS Patna.
  - Establishing Integrated Muscular Dystrophy Rehabilitation Centre "Manav Mandir" at Solan, Himachal Pradesh.
  - Open gymnasium at various locations in Basti Parliamentary Constituency of Uttar Pradesh.
  - Cancer Screening Program of Bihar Govt. and Tata Memorial Cancer Hospital for four Districts of Bihar.
  - Conducting Jan Arogyam Community Healthcare Programme in Nuh District (Aspirational District) of Haryana through Bisnouli Sarvodaya Gramodyog Sewa Sansthan (BSGSS).
  - Support for Electro Chemotherapy Device & Electrodes for use at All India Institute of Medical Sciences, New Delhi.
  - Support to Capital Hospital, Bhubaneshwar, Odisha for Medical Equipment for Eye department Support for construction of Eye Hospital at Dadri, Gautam Buddha Nagar, UP.
  - Support committed for setting up Tele-Recording Room at AIIMS, New Delhi.
- Your Company is operating Directly Observed Treatment cum Designated Microscopy Centre (DOTs cum DMC) at 11 locations of your Company. The program supports the Government of India's National Tuberculosis Elimination Program (NTEP) to prevent, control and eventually eliminate tuberculosis.

Sanitation and Hygiene

- Your Company supported the Government of India's Swachh Bharat Abhiyan or Swachh Bharat Mission, a nation-wide cleanliness campaign by the Government of India, by making available more than 26,000 toilets in about 16000 schools covering 650 blocks in 83 districts spread over 17 states.

- Your Company has also supported for Pilot Project for Mechanized Cleaning of 14 wards in Varanasi.
- Your Company has revived the defunct Municipal Solid Waste Processing Plant at Karsada, Varanasi. Your Company is also operating and maintaining the plant since July' 2017 onwards.
- Your Company with active involvement of its employees, local school children etc. had organises Nukkad Natak, debates, slogan competitions, essay competitions, painting competitions, walkathons, pamphlet distribution etc. in the vicinity of its various projects and stations to spread awareness about sanitation and hygiene.
- Your Company has constructed Individual Household Toilets and Public Toilets in the adjoining villages and Public Places located in the vicinity of our Projects and Stations with an objective of creating an open defecation free society.
- Your Company is supporting the developmental & beautification works of Charminar, Hyderabad, under 'Swachh Iconic Places Project' of Govt. of India. The goal of the Initiative is to improve the cleanliness conditions of places that are "iconic" due to their heritage, religious and/or cultural significance.
- Your Company has encouraged the formation of women's self-help groups to produce and distribute low-cost sanitary napkins in the villages surrounding its power plants.
- Your Company has committed to support the setting up electric crematorium in Noida.

Water

- Your Company ensures access to water to community through installation of hand pumps and piped drinking water etc. To ensure that people have access to potable drinking water, your Company has set up RO water plants/ Water ATMs in public locations near your Company operations. Your Company also distributes water filters/ coolers in various villages/ schools near your Company operations. Further, during extreme summers, your Company ensures availability of water through Water Booths and Water Tankers.
- Your Company has taken initiatives of rejuvenation of Ponds located in the vicinity of many of its Plants with an objective to improve ground water table.
- Your Company has Installed 06 nos. e-Pios (RO systems) at various hospitals in Balasore, Odisha.
- Your Company has extended support for installation of about 10,000 Energy Efficient Pump System in the fields of farmers residing near your Company stations located in 05 districts of UP.

Your Company's CSR activities in other areas

Skill Development and Income Generation

- Your Company has adopted 18 nos. ITIs and set up 8 new ITIs at various locations.
- Your Company is supporting "Skill India Mission" of GoI through MoU with NSDC for various employment linked skill development programs for 30000 rural youth including 8,000 youth of J&K.
- Your Company has provided various capacity building training programmes, exposure visits and provided hand holding through experts to the farmers of villages in its vicinity on various techniques of crop/animal productivity such as use of drip irrigation to produce more crop per drop, in improving milk yield through improving breed through artificial insemination, cultivation of nutrient rich and rapidly growing fodder crops etc.
- Your Company makes youth entrepreneurial, enterprising and employable by providing them with training in Electrical Repairing, Mobile Repairing, Motor Rewinding, Welding, Car Driving including obtaining LMV driving license, Computer Training, etc.

Women Empowerment

- Your Company has established 50 Sanitary Napkin Mini Manufacturing Units (MMU) in the state of Odisha under "Stree Swabhiman", an initiative for providing affordable and accessible sanitary products close to the homes of adolescent women and girls in rural areas.
- Your Company has supported Udyan Care at Jaipur, Rajasthan for the higher education and rehabilitation of orphan and abandoned girls.
- Your Company has conducted various training on embroidery, dress designing, cutting, stitching, tailoring (including providing sewing machines), beautician, food preservation & processing, nursing etc. to women from various villages located in its vicinity.

Support for Physically Challenged Persons

- In collaboration with National Institute for Locomotor Disabilities, your Company has set up Disability Rehabilitation Centers (DRCs) at Bongaigaon, Farakka, Dadri, Korba, Tanda and Rihand for mainstreaming of physically challenged persons (Divyangjan) by providing them with aids and appliances and restorative surgery.
- Information and Communication Technology (ICT) centers established at Delhi University, Guwahati University and Devi Ahilya Vishwavidyalaya, Indore equipped with latest adaptive and assistive technologies so that persons with disabilities can perform equally well in the classroom and workplace.
- Your Company distributes equipment and appliances and takes up activities like inclusive education & vocational training benefitting PCPs all across locations of your Company.

Sports

- Your Company has adopted Archery as a sport with the objective of scouting for talent in remote parts of India and nurture them through coaching camps to enhance India's presence in the sports internationally. The 3-year agreement with Archery Association of India (AAI) includes the title rights across all National Archery Championships (NACs), National Ranking Archery Tournaments (NRATs) and exclusive right to provide kits/ apparels to Indian Archery team for international participation.
- Your Company supported Aizawl Football Club in Aizawl, Mizoram by providing 02 buses.
- Your Company's support is committed for creating additional facilities such as swimming pool in "Multipurpose Sports Complex" under construction at Dharwad, Karnataka.

Rural Infrastructure Development

- Your Company supports the development of basic infrastructure such as roads, bridges, culverts, bus shelters, community centers, schools, health centers enabling the local community to fulfil their basic needs and to enhance the quality of their lives.
- Installation of LED based Solar Street Lights & Solar High Mast lights in Sant Kabir Nagar, Uttar Pradesh and in Satna, Madhya Pradesh.
- Your Company supported the development of Brahma Jahari Forest in Chaumuha in Mathura district, UP.
- Your Company's support is committed for construction and redevelopment of Shri Badrinath Dham town as a spiritual smart hill town.

Preservation of Heritage

- Your Company has supported Archeological Survey of India (ASI) and National Culture Fund (NCF) for preservation and conservation of 3 monuments: Group of Monuments, Mandu (MP), Excavated site at Vikramshila (Bihar) and Archaeological site, Lalitgiri/ Dhauli (Odisha).
- Support to Society for Development of Rural Literature under your Company CSR for Purvasha Folk and Tribal Art Museum, near Chilika Lake, Odisha for protection and promotion of heritage of dying art forms of Odisha.

Relief during Natural Calamity

- Financial support to Uttarakhand State Disaster Management Authority (USDMA) for undertaking reconstruction and restoration of Govt. Schools and Govt. Health Centers in various districts of Uttarakhand.
- Your Company is supporting the Redevelopment of Kedarnath town, Uttarakhand and its surrounding areas devastated during natural calamity of 2013.

5. **Have you taken steps to ensure that this community development initiative is successfully adopted by the community? Please explain in 50 words or so.**

Community Development initiatives in your Company are taken in a planned way. The community is engaged in all steps of activities from planning to completion.

Bottom up approach is adopted for taking up these activities. CSR activities/ CD plans are identified after consultations with relevant stakeholders like community, village panchayats, local/ district administration & Village Development Advisory Committee based on Need Assessment Surveys.

Stakeholders participate at each stage of the activity and also execution of some of the activities is supervised by the local authorities. People's involvement is also ensured during implementation and monitoring.

Your Company's flexible & open approach in this regard coupled with community participation along with local administration & village Panchayats leads to successful adoption & acceptability of initiatives by community. Assets are handed over to local authorities and gram panchayats for maintenance.

Social Impact Evaluation (SIE) study/ survey are conducted through external agencies on completion of developmental initiatives. Social audit is done from time to time to evaluate whether all activities have been completed satisfactorily that gives recommendation for necessary modification/ corrective measures, if any, for the future projects.

Principle 9: Businesses should engage with and provide value to their customers and consumers in a responsible manner

1. **Customer complaints / consumer cases are pending as on the end of financial year:**

   There are no complaints as such from customers. However, as part of the tariff determination process under the overall regulatory system, different cases have been filed by your Company against CERC/ Customers or filed at the Appellate Tribunal/ Courts by different Beneficiaries against your Company/ CERC. The total number of such cases presently pending in APTEL and other courts is 100 as on 1.4.2022.

2. **Does the company display product information on the product label, over and above what is mandated as per local laws?**

   Not Applicable

3. **Is there any case filed by any stakeholder against the company regarding unfair trade practices, irresponsible advertising and / or anti-competitive behaviour during the last five years and pending as on end of financial year? If so, provide details thereof, in about 50 words or so.**

   Nil

4 **Did your Company carry out any consumer survey/ consumer satisfaction trends?**

Yes.

As part of the Customer Relationship Management programme, your Company conducts a Customer Satisfaction Survey to assess the satisfaction level of the customers, captured through an Index and to get feedback from the customers.

Customer satisfaction surveys conducted in the past five years and targets for next 5 years are tabulated below:

Particulars	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
% of satisfied customers	95.2	96.1	95.5	97	96.56	100	100	100	100	100
Coverage %	100	100	100	100	100	100	100	100	100	100

For and on behalf of the Board of Directors

(Gurdeep Singh)
Chairman & Managing Director

Place: New Delhi
Dated: 30th July, 2022